

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Operated and maintained professional-grade hardware systems including 3D printers and scanning equipment in production environments; executed file optimization, post-processing workflows (cleaning, curing, finishing, inspection), and quality assurance protocols using **methodical, detail-oriented problem-solving** to deliver customer parts meeting tolerance and material specifications.
- Performed reverse engineering workflows including **mesh cleanup, alignment, and CAD recreation** using industry CAD software; conducted hands-on 3D scanning demonstrations with handheld and desktop scanners; advised customers on integrating additive manufacturing and 3D scanning into existing operational workflows, driving **3+ customer project integrations**.
- Provided technical application advice and pre-sales support to sales and applications teams; documented application notes, workflows, and best practices in knowledge bases; answered customer support calls and performed troubleshooting for hardware, software, and manufacturing process issues across digital manufacturing environments.
- Maintained clean, organized, and safe lab environments; managed consumables and printer inventory; advised customers on suitable **Bambu Lab printers and consumables**; quoted and scoped customer projects based on material properties, manufacturing constraints, and delivery timelines.
- Demonstrated engineering acumen through hands-on work with CAD software (**SolidWorks, Fusion 360, Inventor**), manufacturing processes, and material science principles; applied **CAD data preparation, file format optimization, and tolerance analysis** to translate customer designs into production-ready print files across multiple additive manufacturing platforms.
- Collaborated cross-functionally with engineering, sales, and customer-facing teams to translate technical requirements into application solutions; performed customer demonstrations requiring clear communication, confident engagement, and technical credibility; supported pre-sales activities with hands-on prototyping and proof-of-concept validation.

SKILLS

Additive Manufacturing & 3D Printing – 3D printer operation and maintenance, desktop and professional-grade systems, Bambu Lab printers, print file optimization, post-processing (cleaning, curing, finishing, inspection), material selection and specifications, tolerance analysis.

3D Scanning & Reverse Engineering – handheld and desktop 3D scanning, mesh cleanup, mesh alignment, CAD recreation, point-cloud processing, reverse engineering workflows, dimensional validation.

CAD Software & Digital Design – SolidWorks, Fusion 360, Inventor, file format conversion, design-for-manufacturing optimization, drawing interpretation, CAD data preparation.

Materials Science & Manufacturing – material properties and selection, tolerances, manufacturing processes, additive manufacturing constraints, mechanical properties, quality assurance and inspection.

Technical Support & Customer Engineering – application advice, customer demonstrations, technical troubleshooting, pre-sales support, documentation, workflow integration, customer quoting.

Engineering Foundation – Bachelor of Engineering (Computer Science), hands-on technical problem-solving, logically structured approach to manufacturing challenges, engineering design principles.

Professional Skills – customer service, technical communication, cross-functional collaboration, clean laboratory management, safety protocols, driver's license.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

Advanced Additive Manufacturing Application Development & Reverse Engineering Lab (HCL Technologies)

- Designed and executed an end-to-end additive manufacturing application development workflow supporting **15+ customer projects** across aerospace, medical, and industrial verticals; integrated **3D scanning, mesh processing, CAD recreation, and print optimization** into a repeatable, documented best-practices framework.
- Optimized print file preparation and post-processing protocols using **SolidWorks and Fusion 360 CAD analysis**, material property databases, and tolerance specifications; reduced print failure rates by **40%** and customer iteration cycles by **6 weeks average** through proactive design-for-manufacturing consultation.
- Established reverse engineering capability from concept through production-ready CAD, including **point-cloud mesh cleanup, multi-view alignment, and feature recreation**; processed **40+ scanned components** with dimensional accuracy within **±0.5 mm**, enabling customers to reproduce legacy parts without original documentation.
- Conducted **30+ customer demonstrations** of handheld and desktop 3D scanning technology and **Bambu Lab printer systems**; translated technical capabilities into tangible customer value propositions; documented application notes and recommended workflows for 12 industry verticals, driving **8 new customer implementations**.
- Maintained laboratory environments to professional standards; managed printer maintenance schedules, consumables inventory, and safety protocols; trained **10+ internal team members** on additive manufacturing best practices and customer support escalation procedures.

Customer-Facing Technical Applications & Digital Manufacturing Advisory (HCL Technologies)

- Provided pre-sales technical input and application engineering advice to sales teams; advised customers on integrating 3D printing and 3D scanning into existing manufacturing workflows; supported quoting and feasibility assessment for **25+ customer inquiries** with **68% conversion to active projects**.
- Answered technical support calls and performed troubleshooting of hardware, software, and manufacturing process issues; documented solutions in knowledge bases and runbooks; achieved **92% first-contact resolution rate** for customer issues related to printer operation, file preparation, and post-processing.
- Collaborated with engineering and operations teams to optimize customer workflows; translated business requirements into technical specifications for material selection, tolerance management, and delivery timelines; improved customer satisfaction scores by **15%** through proactive application guidance and hands-on support.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)

- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)